

|           |                                                                                                                                                                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | not exactly equal to the difference between the energy levels.                                                                                                                                                                                          |
| <b>29</b> | $\Delta x \Delta p \geq h$ <p>Since <math>\Delta x = 3.0 \times 10^{-10} \text{ m}</math></p> $\Delta p \geq \frac{h}{3.0 \times 10^{-10}} = \frac{6.63 \times 10^{-34}}{3.0 \times 10^{-10}}$ $\Delta p \geq 2.21 \times 10^{-24} \text{ kg m s}^{-2}$ |
| <b>30</b> | $\text{KE} = qV = hc/\lambda_{\text{min}}$ <p>If <math>V</math> is reduced by 1% <math>\lambda_{\text{min}}</math> is increased to 2 times (100.0%)</p>                                                                                                 |